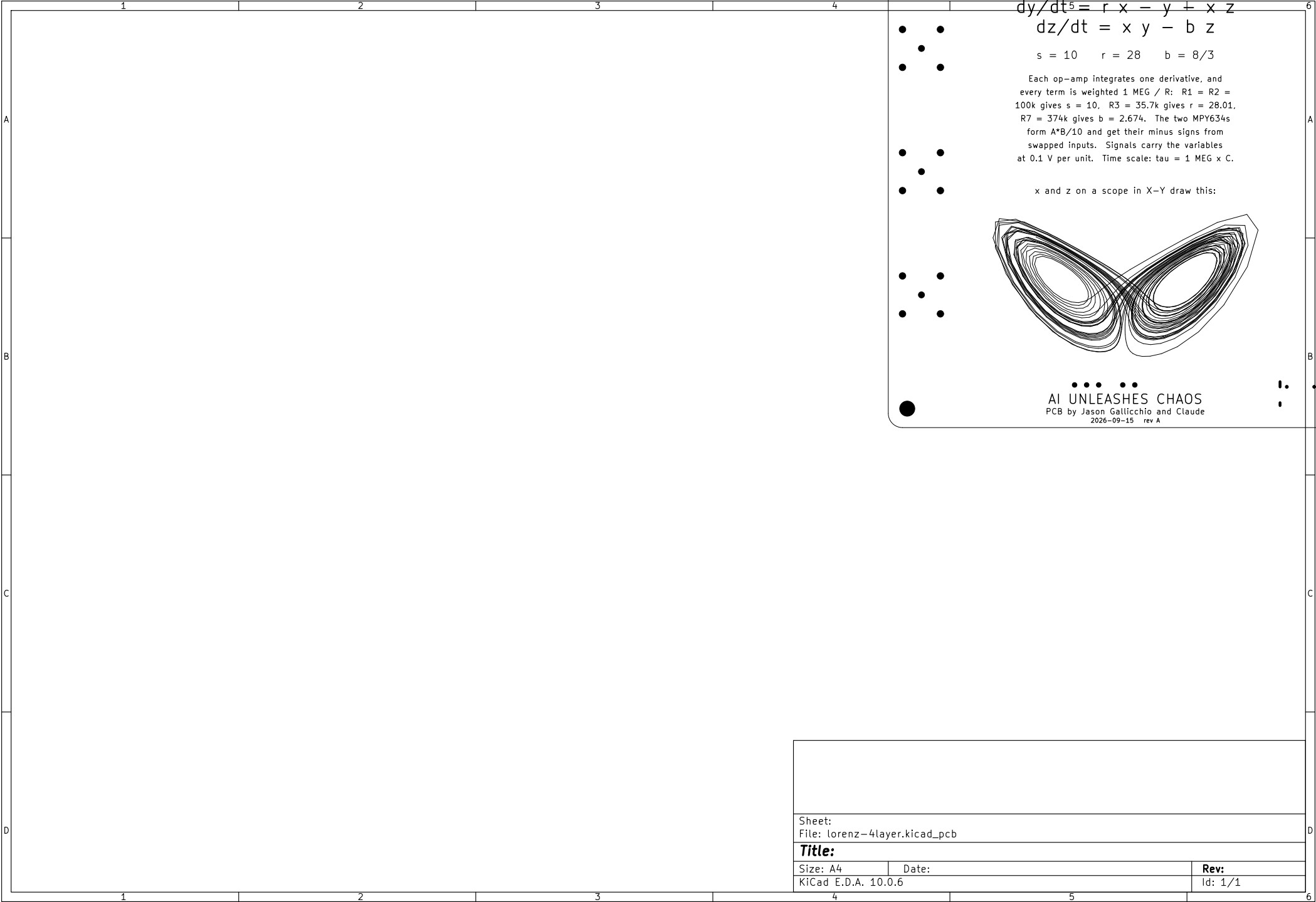


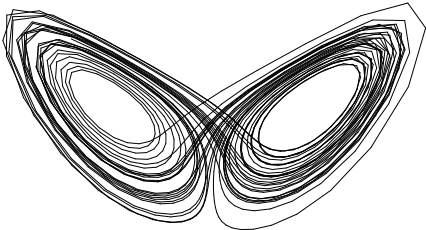


$$\begin{aligned} \frac{dx}{dt} &= s (y - x) \\ \frac{dy}{dt} &= r x - y + x z \\ \frac{dz}{dt} &= x y - b z \end{aligned}$$

$$s = 10 \quad r = 28 \quad b = 8/3$$

Each op-amp integrates one derivative, and every term is weighted 1 MEG / R: R1 = R2 = 100k gives s = 10, R3 = 35.7k gives r = 28.01, R7 = 374k gives b = 2.674. The two MPY634s form A*B/10 and get their minus signs from swapped inputs. Signals carry the variables at 0.1 V per unit. Time scale: tau = 1 MEG x C.

x and z on a scope in X-Y draw this:



AI UNLEASHES CHAOS
PCB by Jason Gallicchio and Claude
2026-09-15 rev A

Sheet:		
File: lorenz-4layer.kicad_pcb		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 1/1